
BPhO Computational Physics Challenge 2026

Ten computational models and interactive extensions

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Project links

[Source code repository](#) • [Interactive photoelectric simulator](#) • [Project video](#)

Listing order of names is random; equal contribution.

Oliver completed Questions 1 to 5 plus the extension.

Richard completed Questions 6–10 (Questions 6 to 10 being the harder set).

Abstract

This project implements all ten tasks in the BPhO Computational Physics Challenge using MATLAB, together with a browser-based photoelectric-effect experiment and molecular-orbital extensions to the hydrogenic-orbital task. The programs progress from stochastic motion to thermal and quantum models, with an emphasis on interactivity, physical interpretation and numerical self-consistency checks. Each section below states the governing model, summarises its implementation and records the principal validation or limiting behaviour. The complete source code, rather than long listings, is supplied in the linked repository.

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1 Project overview

The ten tasks were implemented as independent, reproducible programs. Tasks 1–5 use MATLAB scripts or graphical applications developed primarily by Oliver Yang, while Tasks 6–10 use a common interactive MATLAB layout developed primarily by Richard Xu. Oliver Yang additionally developed the static website for Task 4 and the H_2^+ and H_3^+ linear-combination-of-atomic-orbitals (LCAO) extensions. Both authors tested the models, reviewed the physics and prepared the presentation. This statement should be edited if the final division of work differs.

The general workflow was:

1. express the supplied physical model in consistent units;
2. evaluate it numerically over a suitable parameter range;
3. expose important parameters through sliders, selectors or editable constants;
4. visualise the result in a form that changes immediately with the model;
5. compare against an analytical identity, limiting case or conservation rule where possible.

The numerical comparisons in this report are described as *self-consistency checks*, rather than experimental verifications, whenever the predicted values are generated from the same equations being checked.

2 Task 1: two-dimensional random walk

For each of N fixed-length steps, an angle is sampled uniformly from $[0, 2\pi)$:

$$\theta_i \sim U(0, 2\pi), \quad \Delta x_i = s \cos \theta_i, \quad \Delta y_i = s \sin \theta_i. \quad (1)$$

The representative path is obtained using cumulative sums. A single walk is not statistically conclusive, so the program also generates 5000 independent walks and records their final displacement R . For large N , the radial distribution approaches

$$p(R) = \frac{2R}{Ns^2} \exp\left(-\frac{R^2}{Ns^2}\right), \quad (2)$$

with

$$\langle R^2 \rangle = Ns^2, \quad R_{\text{rms}} = s\sqrt{N}, \quad \langle R \rangle \simeq \frac{\sqrt{\pi}}{2} s\sqrt{N}. \quad (3)$$

Implementation and check

The final figure combines one trajectory with the ensemble histogram and its large- N Rayleigh prediction. The console reports the percentage difference between the simulated $\langle R^2 \rangle$ and Ns^2 . Coordinates are in arbitrary units because no physical unit is assigned to s .

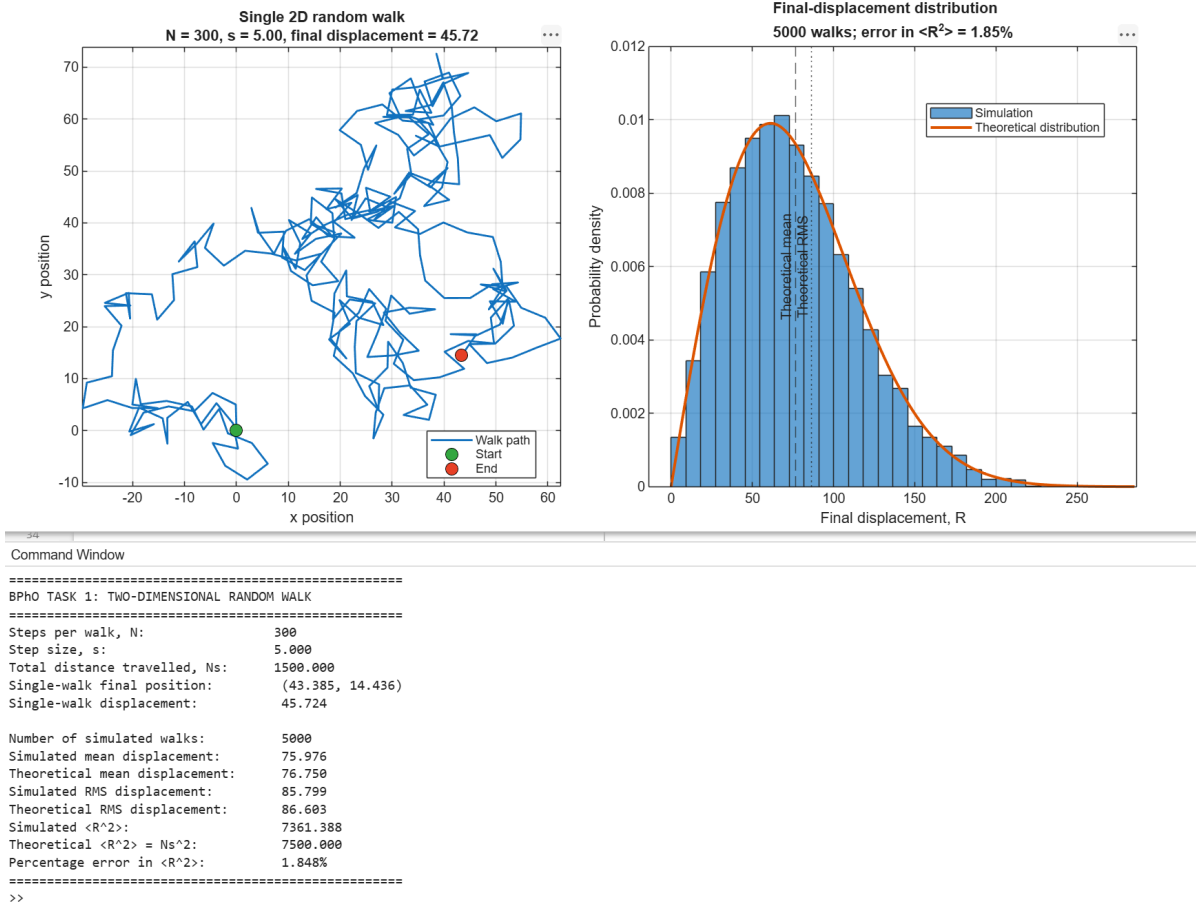


Figure 1: A representative random walk and the ensemble distribution of final displacement.

3 Task 2: Brownian-motion model

The Brownian-motion program places many light hard-disc particles and one much heavier disc inside a reflecting box. Light particles move persistently, with probability

$$P(\text{direction change in } \Delta t) = 1 - e^{-\gamma \Delta t} \quad (4)$$

and a new uniformly random direction when a change occurs. Between collisions, positions are advanced explicitly using

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \mathbf{v}(t)\Delta t. \quad (5)$$

Pairwise elastic collisions update the components of velocity along the line of centres, while positional correction prevents persistent overlap. Repeated small-particle impacts produce the irregular motion of the heavier disc.

The live graph is the squared displacement of one trajectory,

$$R^2(t) = |\mathbf{R}(t) - \mathbf{R}(0)|^2, \quad (6)$$

not an ensemble mean-squared displacement.

Implementation and check

The animation includes reflecting walls, light–light and light–heavy collisions, a trajectory trace and a live $R^2(t)$ plot. For every light–heavy collision the program compares total vector momentum immediately before and after the impulse and records the maximum relative numerical error.

Model scope

Random direction resets provide external stochastic forcing and the walls exchange momentum with the particles. The model is therefore a qualitative mechanical illustration of Brownian motion, not a thermal gas at a specified temperature and not a measurement of a diffusion coefficient.

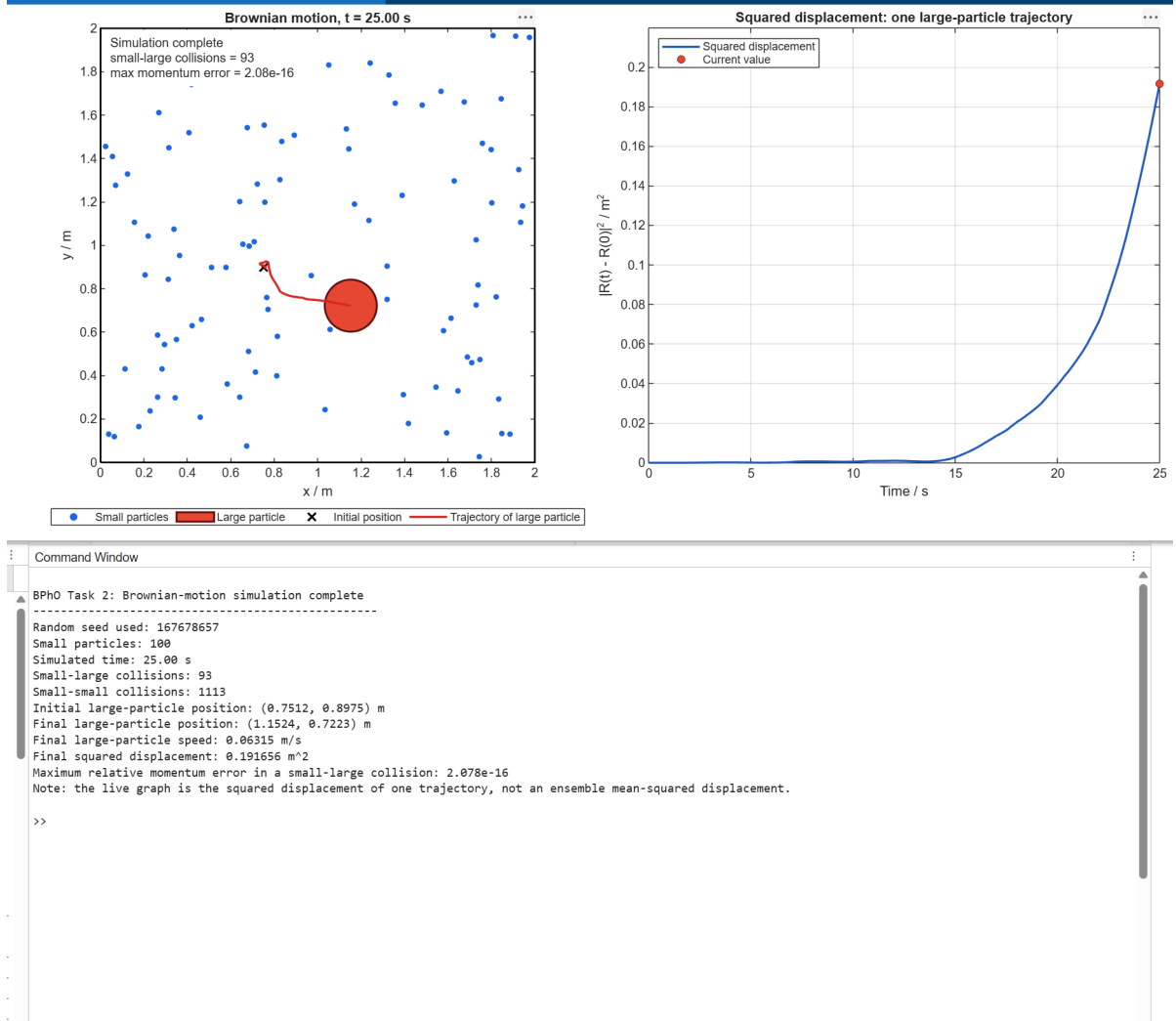


Figure 2: Live Brownian-motion simulation and squared displacement of the heavy particle.

4 Task 3: thermal quantum models

4.1 Planck spectrum and Wien's law

The spectral radiance per unit wavelength is calculated from Planck's law,

$$B_{\lambda}(T) = \frac{2hc^2}{\lambda^5} \frac{1}{\exp(hc/(\lambda k_B T)) - 1}. \quad (7)$$

The program evaluates spectra at several temperatures and finds each numerical maximum. This is compared with Wien's displacement law,

$$\lambda_{\max} T = b, \quad (8)$$

which predicts that the peak moves to shorter wavelengths as temperature increases. The visible interval is shaded to give the scale physical context.

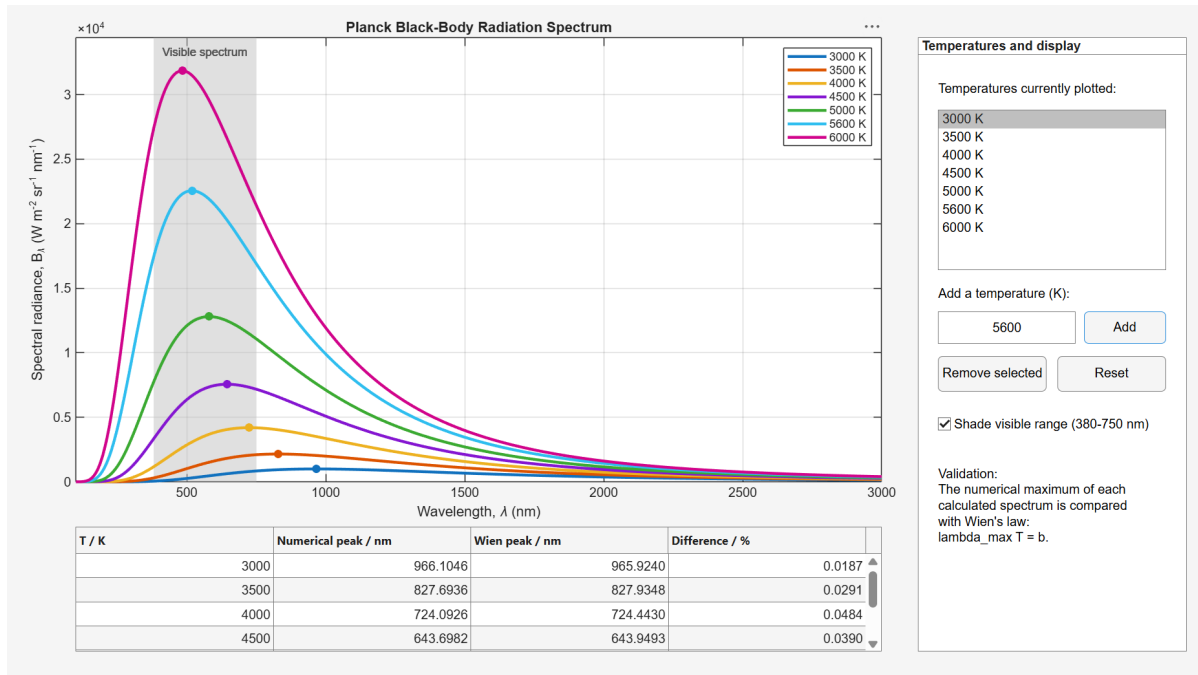


Figure 3: Planck spectra with numerical maxima and Wien-law comparisons.

4.2 Einstein heat capacity

For a solid represented by quantised oscillators, Einstein's molar heat capacity is

$$C_V = 3R x^2 \frac{e^x}{(e^x - 1)^2}, \quad x = \frac{\Theta_E}{T}. \quad (9)$$

At high temperature $x \rightarrow 0$, so

$$C_V \rightarrow 3R, \quad (10)$$

the Dulong–Petit limit. A material with a higher Einstein temperature needs more thermal energy before its vibrational modes approach this classical limit. Gold, copper and iron are displayed by default, while further representative materials may be selected interactively.

Model scope

The plotted curves are predictions of the Einstein model rather than measured data. Additional material parameters should be described as representative unless a source for each Einstein temperature is added.

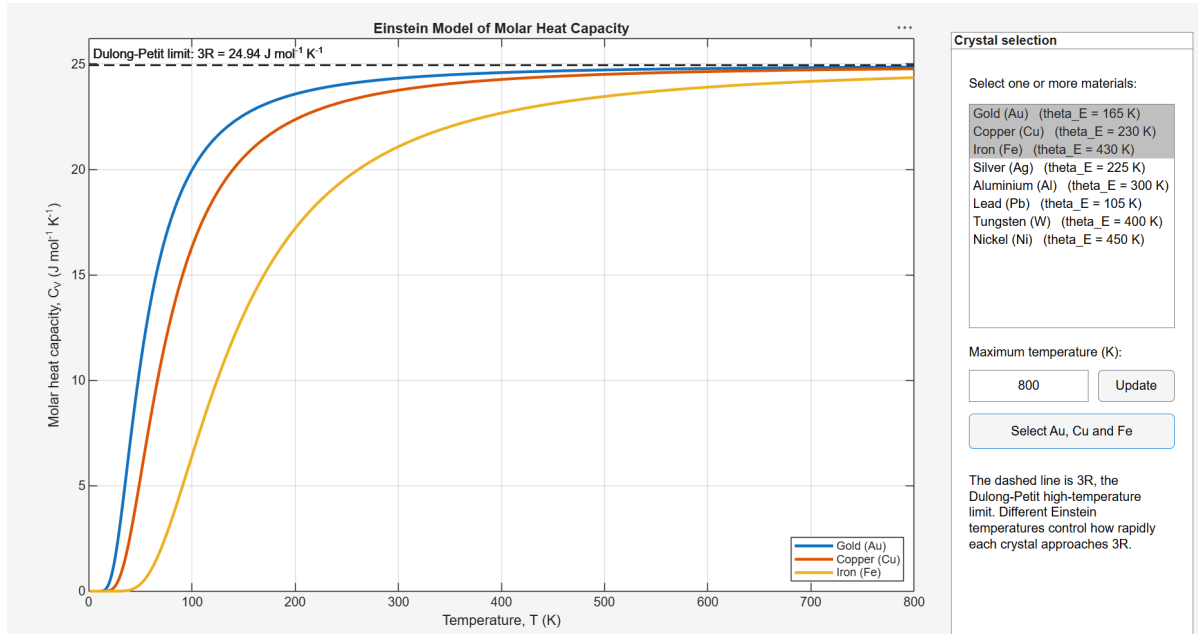


Figure 4: Einstein heat capacities approaching the Dulong–Petit value $3R$.

5 Task 4: photoelectric effect

Einstein’s photoelectric equation gives the maximum photoelectron kinetic energy and stopping voltage:

$$eV_s = hf - \phi, \quad hf \geq \phi. \quad (11)$$

The threshold quantities are

$$f_0 = \frac{\phi}{h}, \quad \lambda_0 = \frac{hc}{\phi}. \quad (12)$$

Thus V_s is linear in frequency with universal gradient h/e , but is nonlinear in wavelength:

$$V_s(f) = \frac{h}{e}f - \frac{\phi}{e}, \quad V_s(\lambda) = \frac{hc}{e\lambda} - \frac{\phi}{e}. \quad (13)$$

The MATLAB application compares five representative metals, marks their thresholds and numerically fits the frequency gradient as an internal check.

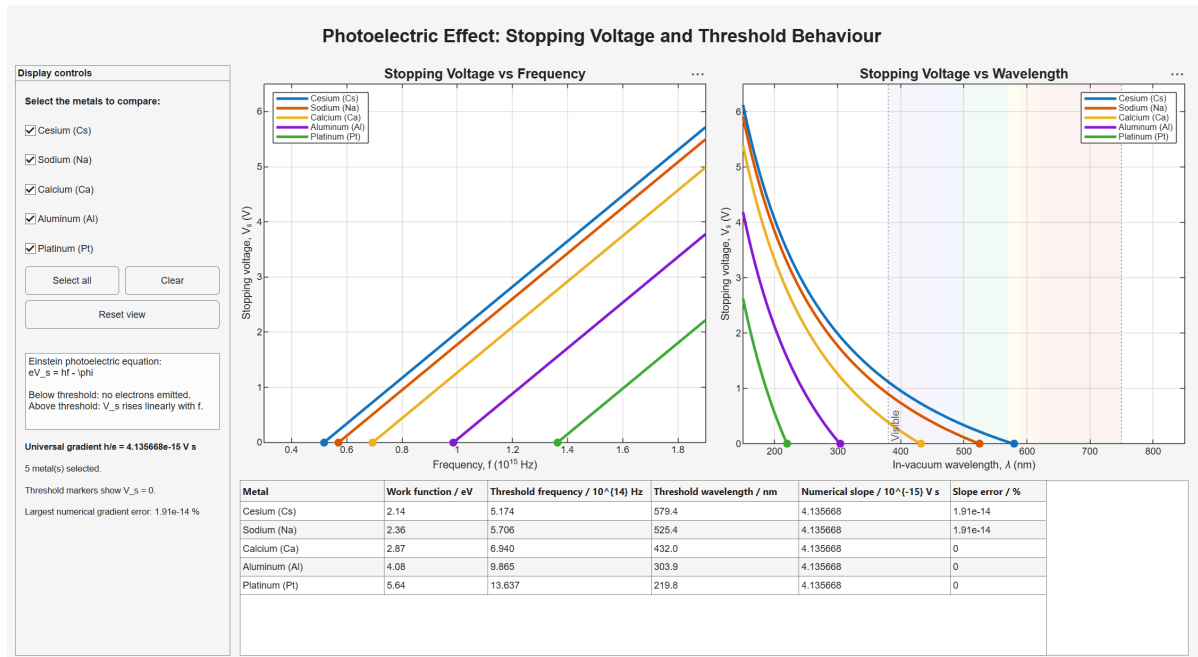


Figure 5: Stopping voltage against frequency and wavelength for several metals.

5.1 Interactive website extension

The static HTML, CSS and JavaScript extension presents a simplified evacuated phototube. The user selects the cathode material, wavelength, intensity and retarding voltage. Each photon has energy

$$E_\gamma = \frac{hc}{\lambda}, \quad K_{\max} = \max(0, E_\gamma - \phi), \quad V_s = \frac{K_{\max}}{e}. \quad (14)$$

Below threshold, no electrons are emitted even if intensity is increased. Above threshold, intensity controls the photon arrival rate and therefore the model photocurrent, but it does not change $K_{\max}$. Increasing retarding voltage reduces the idealised current to zero at V_s . Animated photons and electrons make these distinctions visible without a backend or external API.

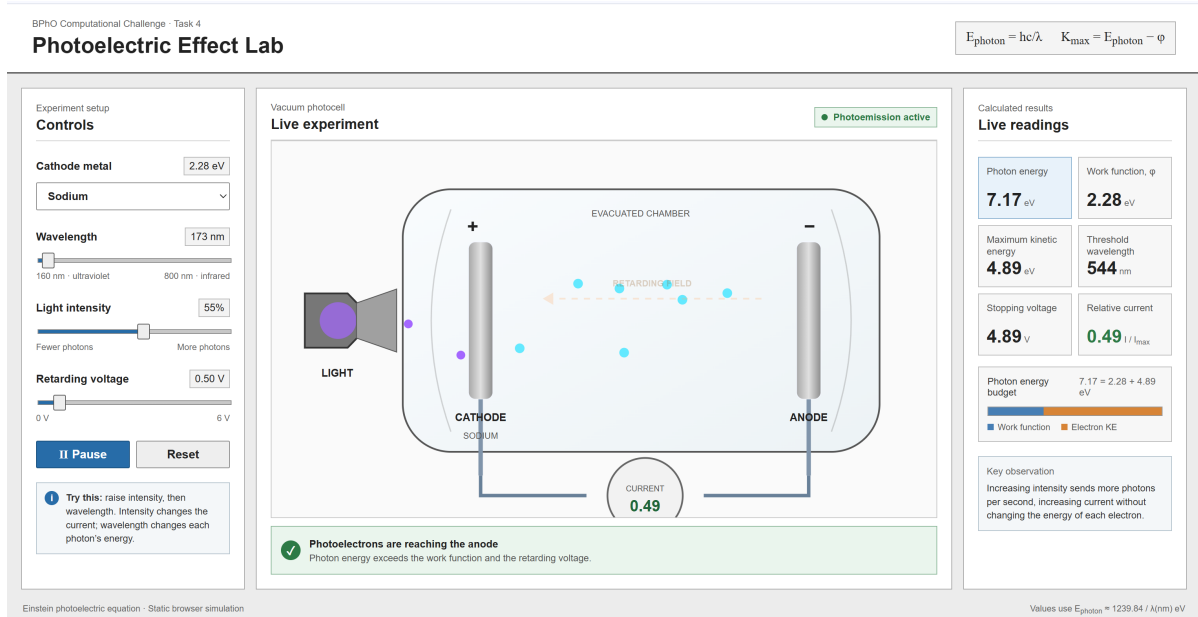


Figure 6: Browser-based photoelectric-effect experiment.

6 Task 5: hydrogen emission spectrum

The Bohr energy levels are represented by

$$E_n = -\frac{13.605693}{n^2} \text{ eV}. \quad (15)$$

For a downward transition $n_i > n_f$, the emitted photon satisfies

$$E_\gamma = E_{n_i} - E_{n_f}, \quad \lambda = \frac{hc}{E_\gamma}. \quad (16)$$

The calculated wavelength is compared with the Rydberg relation

$$\frac{1}{\lambda} = R_\infty \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right). \quad (17)$$

The application displays the Lyman through Pfund series, a logarithmic wavelength axis, visible-region shading and a linked energy-level diagram. Selecting a spectral point highlights the corresponding transition and reports its energy and wavelength.

Implementation and check

The default example is the Balmer H α transition $3 \rightarrow 2$. The agreement between the energy-difference and Rydberg calculations is an algebraic self-consistency check; the model does not calculate line intensity, fine structure or transition probability.

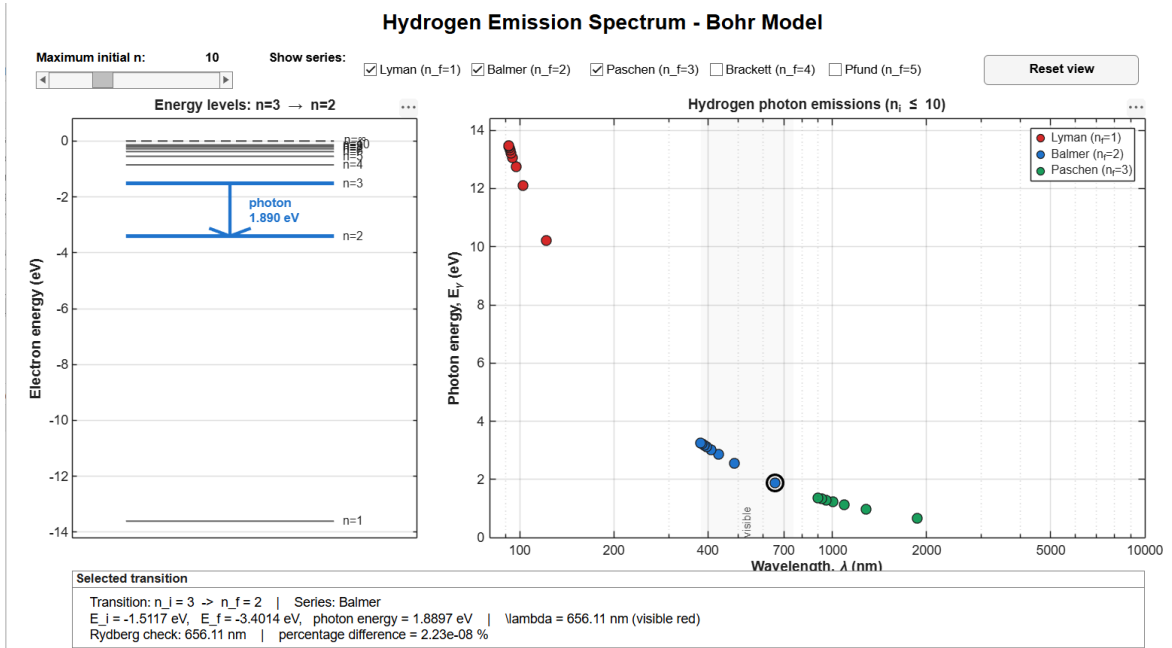


Figure 7: Interactive hydrogen spectrum and linked Bohr energy-level transition.

7 Task 6: electron diffraction

Electrons accelerated through potential difference V have the non-relativistic de Broglie wavelength

$$\lambda = \frac{h}{\sqrt{2m_e eV}}. \quad (18)$$

For first-order diffraction by graphite planes separated by d , Bragg's law gives

$$2d \sin \theta = \lambda. \quad (19)$$

The scattering angle is $\varphi = 2\theta$. On the model spherical screen of radius R , the ring radius is

$$x = R \sin \varphi. \quad (20)$$

Two graphite spacings generate the inner and outer rings. Increasing voltage decreases λ , so both rings contract.

Implementation and check

The program plots $1/\sqrt{V}$ against $\sin(\varphi/2)$ and fits a straight line through the generated points, as required by the combined de Broglie and Bragg relations. This confirms internal consistency, not agreement with independent experimental measurements.

Model scope

The electron wavelength is non-relativistic over the chosen voltage range and the ring brightness is a stylised visual representation, not a calculated scattering intensity.

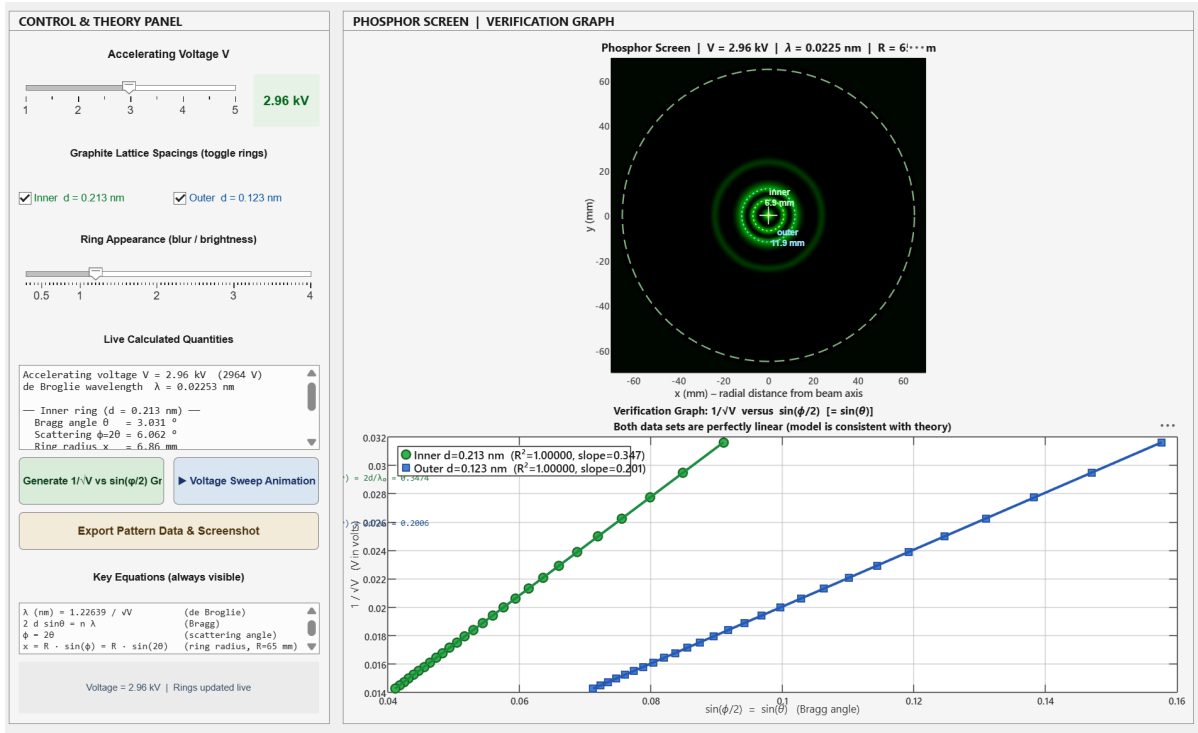


Figure 8: Electron diffraction rings and the linear verification plot.

8 Task 7: particle in an infinite box

For an electron in an infinite one-dimensional well of width L , the normalised eigenstates and energies are

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \quad (21)$$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2m_e L^2}, \quad n = 1, 2, \dots \quad (22)$$

The probability density has $n - 1$ internal nodes. The program varies n and L while updating the energy spectrum and density.

The uncertainty extension uses

$$(\Delta x)^2 = L^2 \left(\frac{1}{12} - \frac{1}{2\pi^2 n^2} \right), \quad (23)$$

$$\Delta p = \frac{n\pi\hbar}{L}, \quad (24)$$

$$\frac{\Delta x \Delta p}{\hbar} = \sqrt{\frac{n^2 \pi^2}{12} - \frac{1}{2}} \geq \frac{1}{2}. \quad (25)$$

Implementation and check

The interface highlights the selected state on the $E_n \propto n^2$ spectrum, plots $|\psi_n|^2$ and compares $\Delta x \Delta p$ with $\hbar/2$ for each allowed n .

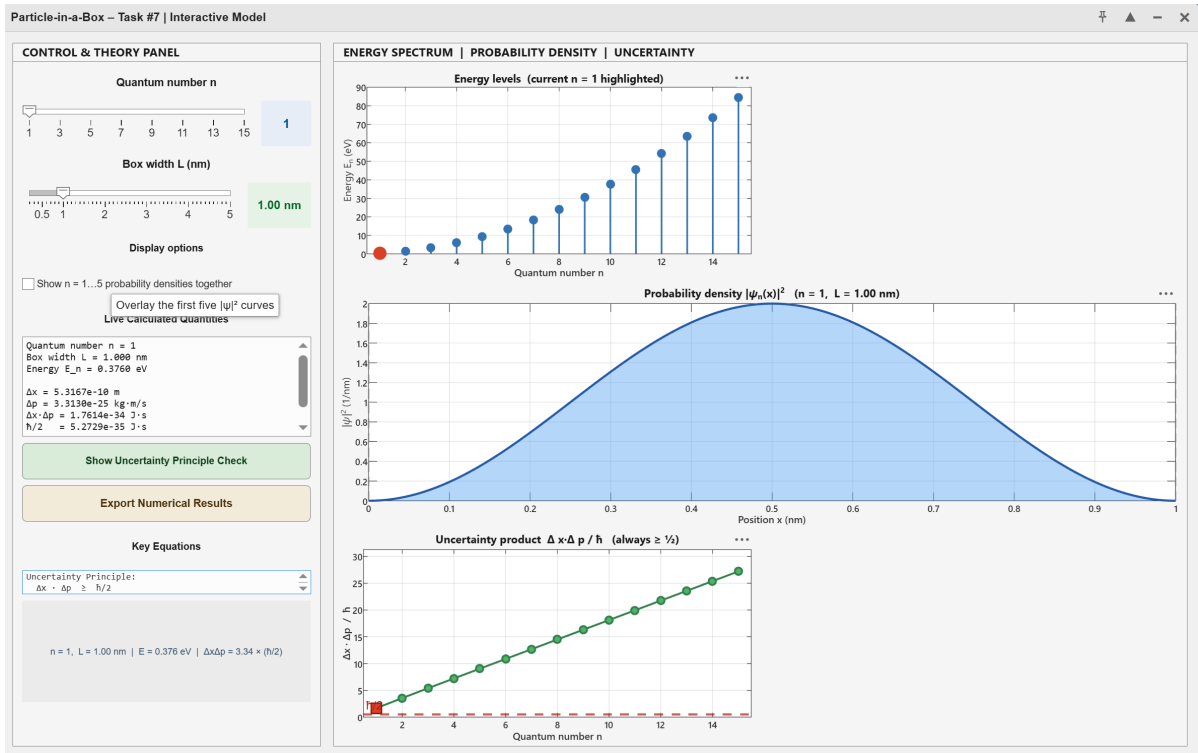


Figure 9: Energy levels, probability density and uncertainty product for an infinite well.

9 Task 8: quantum mismatch probabilities

The model considers the polarisation singlet state

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|HV\rangle - |VH\rangle). \quad (26)$$

Let α be the folded relative detector angle. Identical detector orientations give opposite raw outcomes. If Bob flips his anticorrelated bit before comparing a key, a consistent mismatch comparison is

$$P_Q(\alpha) = \sin^2 \alpha, \quad P_C(\alpha) = \frac{2|\alpha|}{\pi}, \quad |\alpha| \leq \frac{\pi}{2}. \quad (27)$$

Sliders vary Alice's and Bob's analyser directions while a schematic and both probability curves update in real time.

Model scope

These two curves illustrate differing angular correlations. They do not, by themselves, evaluate a CHSH statistic, prove Bell violation or constitute a complete security analysis of a cryptographic protocol.

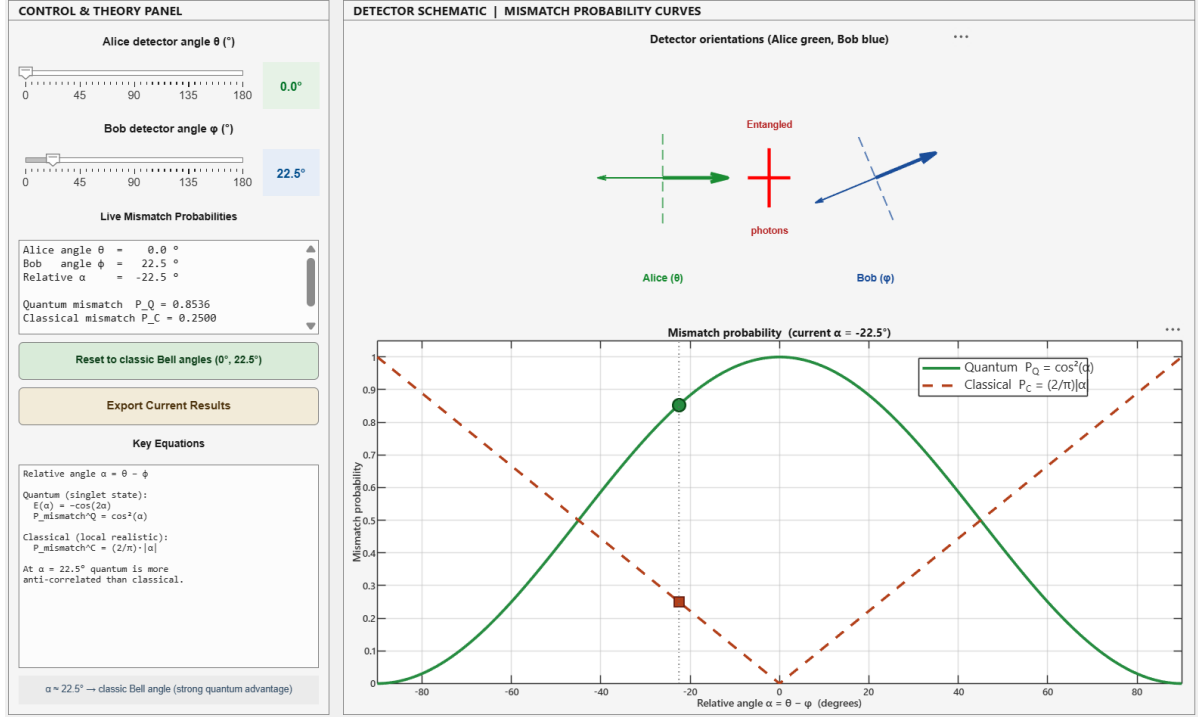


Figure 10: Classical and quantum post-flip key-mismatch predictions.

10 Task 9: Compton scattering

For a photon scattering from a free electron initially at rest, the wavelength shift is

$$\Delta\lambda = \lambda_C(1 - \cos\theta), \quad \lambda_C = \frac{h}{m_e c}. \quad (28)$$

The scattered photon energy and electron kinetic energy are

$$E' = \frac{hc}{\lambda + \Delta\lambda}, \quad K = E - E'. \quad (29)$$

The recoil speed is calculated relativistically using

$$\gamma = 1 + \frac{K}{m_e c^2}, \quad \frac{v}{c} = \sqrt{1 - \frac{1}{\gamma^2}}, \quad (30)$$

and momentum conservation gives the electron recoil direction through

$$\cot\phi = (1 + \alpha) \tan\frac{\theta}{2}, \quad \alpha = \frac{\lambda_C}{\lambda}. \quad (31)$$

Limiting checks

At $\theta = 0$, the wavelength shift is zero. At $\theta = \pi$, the shift reaches $2\lambda_C$. The interface updates markers on plots of fractional shift, v/c and recoil angle as θ and the incident wavelength change.

Model scope

The model assumes a free stationary electron and does not calculate a scattering cross-section. The recoil direction is undefined in the exact zero-scattering limit, where the program must display a conventional value.

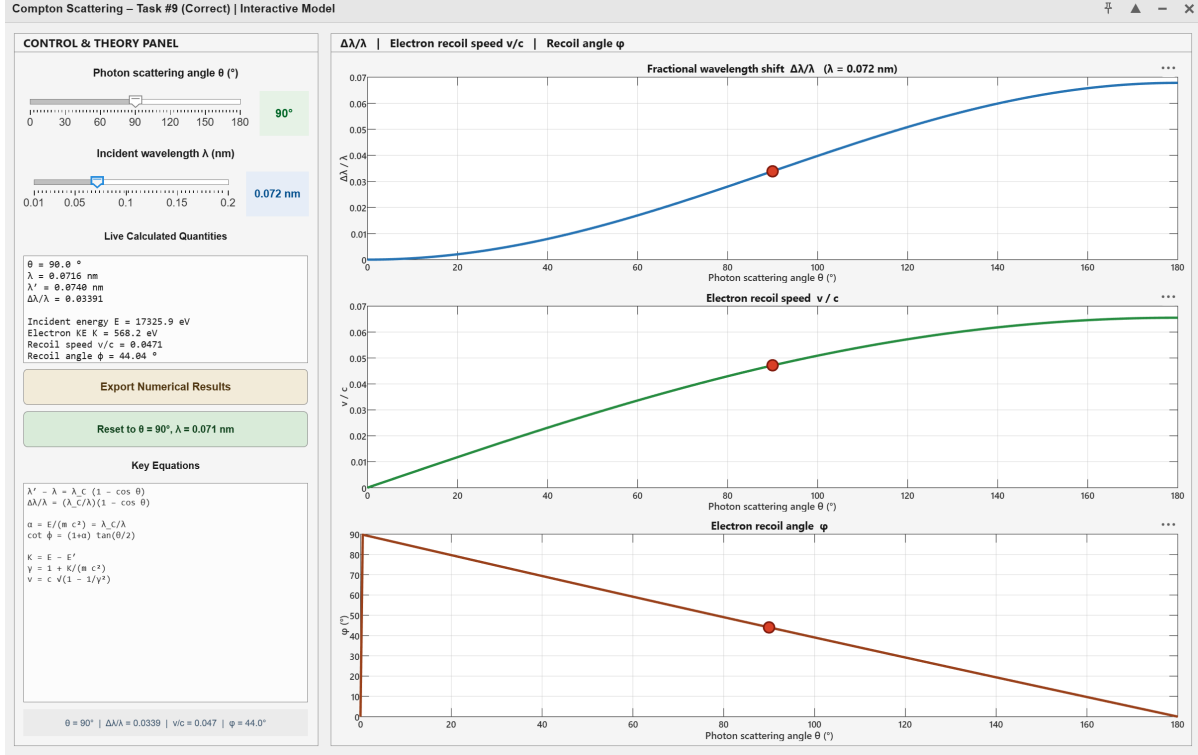


Figure 11: Compton wavelength shift, electron recoil speed and recoil direction.

11 Task 10: hydrogenic orbitals

For a one-electron hydrogen-like ion, separation in spherical coordinates gives

$$\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_{\ell}^m(\theta, \phi), \quad E_n = -\frac{13.605693 Z^2}{n^2} \text{ eV}. \quad (32)$$

Allowed quantum numbers satisfy

$$n \geq 1, \quad 0 \leq \ell \leq n - 1, \quad -\ell \leq m \leq \ell. \quad (33)$$

The interface enforces these ranges and displays a two-dimensional slice of $|\psi|^2$ beside a three-dimensional density isosurface. Complex m eigenstates colour the surface by phase; real combinations colour it by the sign of ψ . Changing Z contracts the orbital and changes the Coulomb energy while retaining the same angular node structure.

Numerical checks

The program checks normalisation, reports the expected $n - \ell - 1$ radial nodes and ℓ angular nodes, and adapts the plotted radius so that the radial grid contains essentially all of the probability. The Coulomb energy depends on n and Z , but not on ℓ or m in this non-relativistic model.

Model scope

$Z > 1$ represents one-electron ions such as He^+ , not neutral multi-electron atoms. The chosen isosurface is a percentage of the sampled maximum density; it is a visual level, not a surface containing a fixed probability, and may suppress weak outer lobes.

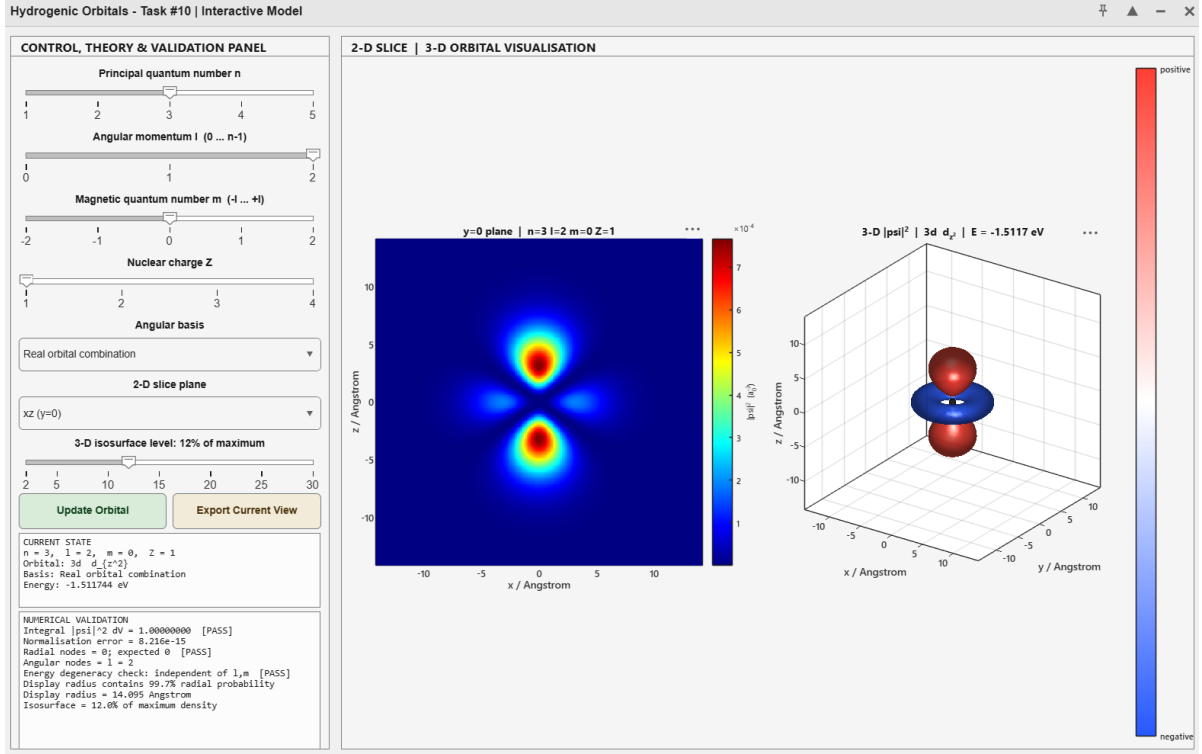


Figure 12: Two- and three-dimensional views of a hydrogenic orbital.

12 Extension: molecular orbitals from LCAO

12.1 H_2^+ : two-centre combinations

Two normalised hydrogen $1s$ orbitals, ϕ_A and ϕ_B , are centred a distance R apart. The bonding and antibonding combinations are

$$\psi_+ = N_+(\phi_A + \phi_B), \quad \psi_- = N_-(\phi_A - \phi_B). \quad (34)$$

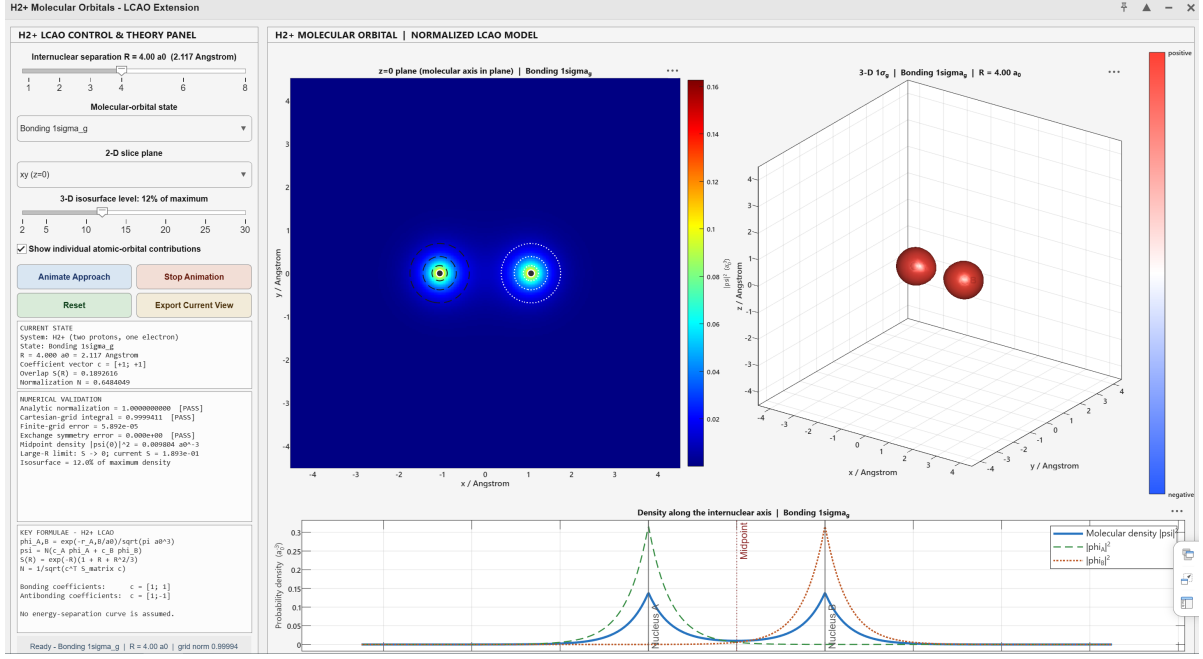
In atomic units, the overlap is

$$S(R) = e^{-R} \left(1 + R + \frac{R^2}{3} \right), \quad (35)$$

so

$$N_+ = \frac{1}{\sqrt{2(1+S)}}, \quad N_- = \frac{1}{\sqrt{2(1-S)}}. \quad (36)$$

The bonding combination has constructive density between the nuclei, whereas the antibonding combination has a midpoint node. The animation varies R to show how the two atomic functions combine as the centres approach.

Figure 13: Normalised two-centre LCAO model for H_2^+ .

12.2 H_3^+ : three-centre combinations

For three $1s$ functions on an equilateral triangle, the minimal molecular orbital is

$$\psi = N \sum_{i=1}^3 c_i \phi_i, \quad N = (\mathbf{c}^\top \mathbf{S} \mathbf{c})^{-1/2}, \quad (37)$$

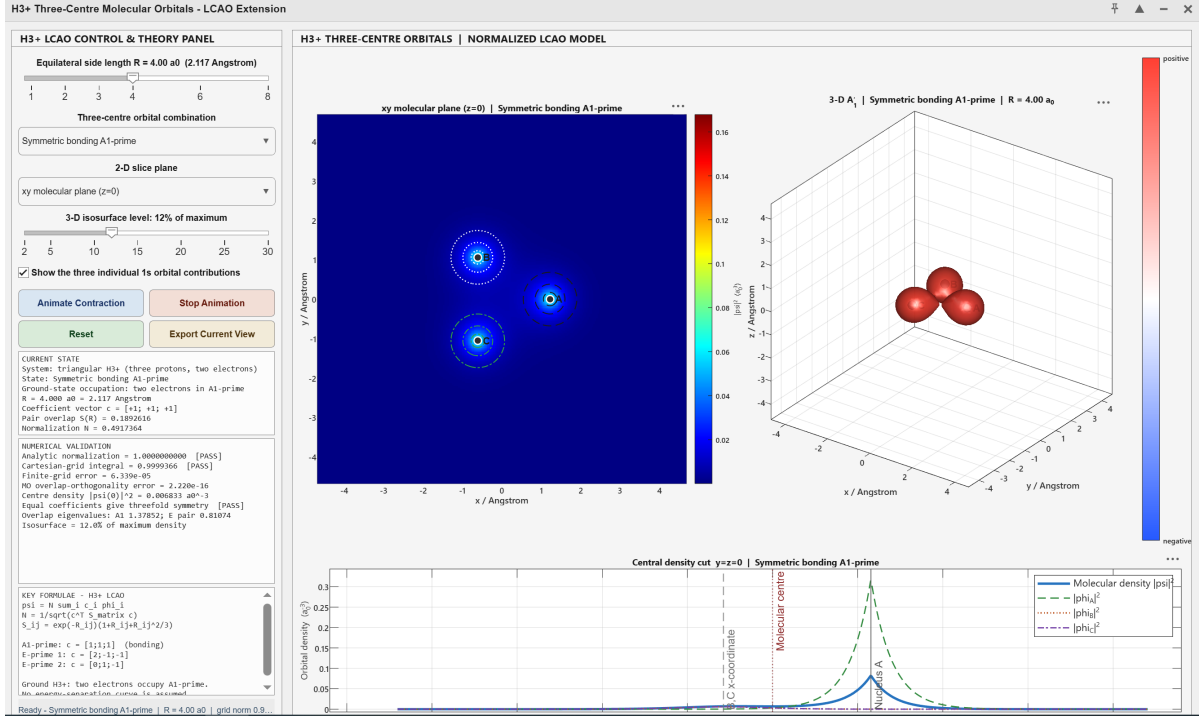
where $S_{ii} = 1$ and every off-diagonal entry is the same two-centre overlap $S(R)$. The symmetric bonding coefficient vector is

$$\mathbf{c}_{A'_1} = (1, 1, 1)^\top. \quad (38)$$

Two components of the nodal E' pair can be represented by

$$\mathbf{c}_{E'_1} = (2, -1, -1)^\top, \quad \mathbf{c}_{E'_2} = (0, 1, -1)^\top. \quad (39)$$

In the minimal picture, the two electrons of the ground configuration occupy the symmetric spatial orbital with opposite spins. The E' combinations are shown to demonstrate how changing coefficients creates nodes.

Figure 14: Normalised three-centre LCAO model for equilateral H_3^+ .

Model scope

The LCAO extensions investigate orbital shape and interference only. They do not compute molecular energies, forces, equilibrium separations or electron correlation. Changing R in an animation is parameter variation, not a time-dependent molecular-dynamics calculation.

13 Summary of numerical checks

Table 1: Principal checks used across the project.

Task	Quantity checked	Expected behaviour
1	Ensemble mean-square displacement	$\langle R^2 \rangle \rightarrow Ns^2$
2	Pair momentum during resolved collisions	Before and after values agree to numerical precision
3	Numerical Planck peak; high- T heat capacity	$\lambda_{\max} T = b$; $C_V \rightarrow 3R$
4	Gradient of $V_s(f)$	$dV_s/df = h/e$
5	Transition wavelength	Energy difference agrees with Rydberg relation
6	Voltage-angle relation	$1/\sqrt{V}$ is linear in $\sin(\varphi/2)$
7	Uncertainty product	$\Delta x \Delta p \geq \hbar/2$
8	Equal detector settings	Zero post-flip key mismatch in the quantum model
9	Forward and backward limits	$\Delta\lambda(0) = 0$, $\Delta\lambda(\pi) = 2\lambda_C$
10	Wavefunction normalisation and nodes	Integral near unity; node counts match n and ℓ
LCAO	Overlap-matrix normalisation and symmetry	$c^T S c$ gives unit norm

14 Conclusion

The project develops from a statistical random walk to collision-driven motion, early quantum models, wavefunctions, scattering and molecular-orbital combinations. The interactive controls were used to make parameter dependence visible, while analytical relations and limiting cases were used to detect implementation errors and distinguish attractive graphics from physically meaningful output. The strongest extensions are the browser-based photoelectric experiment and the normalised two- and three-centre LCAO models.

The models remain intentionally educational. They neglect effects such as full thermalisation, relativistic corrections where unnecessary, fine structure, scattering cross-sections and electron correlation. These limits also identify natural directions for future work.

A Code and reproducibility

The final repository should use MATLAB filenames that match their primary function names. A recommended structure is shown below.

Table 2: Recommended final source filenames.

Component	File
Task 1	Q1_randomwalk.m
Task 2	Q2_Brownianmotion.m
Task 3	task03_thermal_quantum_models.m
Task 4	task04_photoelectric_effect.m
Task 5	task05_hydrogen_emission.m
Task 6	ElectronDiffraction_Task6.m
Task 7	ParticleInABox_Task7.m
Task 8	QuantumCryptography_Task8.m
Task 9	ComptonScattering_Task9_Correct.m
Task 10	HydrogenicOrbitals_Task10.m
LCAO extensions	H2plus_LCAO_Extension.m, H3plus_LCAO_Extension.m
Website	index.html, style.css, script.js

All programs and the final video are available from the links on the title page. Randomised models may use a fixed seed for exact reproduction of a recorded example, while the default unseeded mode demonstrates that the motion is not pre-scripted.

References

- [1] British Physics Olympiad, *Computational Physics Challenge task materials*, 2026. <https://www.bpho.org.uk/>
- [2] D. J. Griffiths and D. F. Schroeter, *Introduction to Quantum Mechanics*, 3rd ed., Cambridge University Press, 2018.
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